# DECISION DEBT OS

## Turning Business Decisions Into Measurable Learning

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Name: Utsav Tyagi  PRN: 26070126233  Branch: AIML  Division: C-3

# OBJECTIVE

• **Specific problem statement:** Businesses make repeated, consequential decisions — retention discounts, pricing changes, feature launches — without recording a measurable assumption about the expected outcome, and without later checking whether that assumption held true.

This creates **Decision Debt** — an accumulation of unresolved assumptions, overdue reviews, invalidated assumptions still driving active work, and repeated forecasting errors. Ordinary dashboards and project-management tools report what happened; none of them capture what a team believed would happen before it acted, so the belief can never be systematically checked.

**Goal:** build a lightweight decision-to-outcome tracking workflow — Decision Debt OS — that converts a business decision into a structured, measurable commitment and automatically evaluates it against reality.

# LITERATURE REVIEW

- Discussing drawback of atleast 3 research papers (existing technology)

| AUTHOR NAME | TITLE | TOOL USED | DRAWBACK | CONCLUSION OF PAPER |
|---|---|---|---|---|
| ScienceDirect — Information & Management | Organisational Memory and Decision Rationale | Literature synthesis / conceptual framework | Describes how organisations store rationale narratively; no forecast, no automated outcome-tracking mechanism | Confirms organisations lose decision rationale over time — motivates structured capture, but offers no tracking system |
| Cambridge — Experimental Economics | Improving Decisions with Market Information: Corporate Prediction Markets | Internal prediction-market experiment | Forecasts a specific question well, but isn't tied to a company's own decision log or retention workflow | Prediction markets measurably improve forecast quality but remain a standalone tool, not a full decision-to-outcome system |
| Springer — J. of Quantitative Economics | Operational Decision Discovery from Data | Process-mining / data-driven discovery | Reconstructs decisions after the fact from logs; cannot capture an assumption that was never logged | Valuable for retrospective analysis, but does not support forward-looking assumption tracking |

*Tool Used / Drawback / Conclusion summarised from each source's stated scope and findings for comparison purposes; full citations on the References slide.*

# NOVELTY

• **What new techniques used:** Combines a churn-risk classifier (Random Forest, trained on real customer-behaviour data) with a rule-based Decision Evaluation Engine that checks a stated forecast against the actual outcome at a deadline, assigning one of six honest validation states — Validated, Partially Validated, Invalidated, Inconclusive, Stale, or Not Measurable.

• **How your topic is different from existing one:** BI dashboards report outcomes only; Jira/Notion capture decisions without a bound forecast; corporate prediction markets (P0, Hypermind) forecast a question but aren't tied to a company's own decision log; enterprise decision-intelligence patents are heavyweight, org-wide systems. Decision Debt OS is the first to combine an explicit, falsifiable assumption with automatic outcome evaluation in a lightweight workflow sized for a single product or retention team.

# DATASET USED

• **Link:** Synthetic dataset generated for this project — no external public-repository link; file supplied directly as Customer_Transactions.csv.

• **Downloaded data set:** 10,000 fictional customer-decision records · 14 fields — age, gender, country, annual income, spending score, number of purchases, average purchase value, membership years, website visits/month, cart-abandon rate, churned (target), feedback text, last purchase date.

**Note:** *clearly labelled synthetic / fictional data, used for prototype demonstration only — not drawn from any real company's customers.*

# ML ALGORITHM

• Implementation of problem formulation using ML

• **Problem formulation:** binary classification — predict whether a customer churns within 30 days (churned = 1) or is retained (churned = 0).

• **Features:** spending score, cart-abandon rate, website visits/month, number of purchases, average purchase value, membership years, annual income, age, gender, country.

• **Preprocessing:** StandardScaler on numeric features, One-Hot Encoding on categorical features (gender, country).

• **Train/test split:** 80% / 20%, stratified by class to preserve the churn ratio.

• **Algorithms compared:** Logistic Regression (baseline) vs. Random Forest Classifier, 300 trees, class_weight="balanced" to correct for churn being only 10.9% of the data.

# IF IMPLEMENTED (RESULT)

• Which tools/language used to implement

**Python 3 · Pandas · NumPy · Scikit-learn · Matplotlib**

**Result (Random Forest, held-out test set, n = 2,000) — real, computed values:**

| 0.26 | 0.78 | 0.39 | 0.84 |
|---|---|---|---|
| Precision | Recall | F1-score | ROC-AUC |

**Interpretation:** optimised for recall over precision — catching 78% of true churners matters more than being right every time, since churn is only 10.9% of the data. The model flags risk; it does not prove any retention action caused a customer to stay — that requires a controlled comparison, which the Decision Evaluation Engine checks for.

# CONCLUSION

• Decision Debt — the accumulation of decisions whose assumptions are never checked against reality — is a real, recurring gap that ordinary dashboards and project trackers do not close.

• Decision Debt OS prototypes a lightweight workflow (Capture → Structure → Forecast → Measure → Evaluate → Learn) that binds a stated assumption to a measurable metric and a deadline.

• On a real, if synthetic, 10,000-row dataset, risk-based targeting is justified: a high-risk cohort churns at 37.2% versus 1.5% for a low-risk cohort, and a Random Forest classifier reaches 0.84 ROC-AUC predicting that risk.

• The model predicts risk; it does not prove causation — human review and a controlled comparison remain required before crediting any intervention.

• Decision Debt OS does not replace human judgment. It makes the assumptions behind human decisions visible, measurable, and reviewable.

# APPLICATIONS

• Customer-retention decisions — discounts, win-back offers, onboarding changes for at-risk customer segments

• Pricing and packaging decisions — testing a price change against a stated, measurable revenue/churn forecast

• Feature-launch decisions — tracking whether a shipped feature met its adoption or retention target

• Growth and marketing experiments — binding a campaign's expected lift to an explicit, checkable number

• Organisation-wide decision registries — a shared log of assumptions, forecasts and outcomes for audit and institutional learning

# REFERENCES


**1.** Organisational Memory and Decision Rationale — ScienceDirect

**2.** Operational Decision Discovery from Data — Springer

**3.** Improving Decisions with Market Information: Corporate Prediction Markets — Cambridge / Experimental Economics

**4.** Incentive-Compatible Forecasting — INFORMS / Management Science

**5.** P0 — Internal Prediction Markets (priorityzero.ai)

**6.** Hypermind — Crowd Forecasting (hypermind.com)

**7.** Vigil — Decision Tracking Use Cases (trustvigil.com)

**8.** Patent EP1402435A4 — Google Patents

**9.** Patent US20250217680A1 — Google Patents